

| <b>RADIONUCLIDE SAFETY DATA SHEET</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                                    |                                                                                                                                                                                                                                                                                                                           |         |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|
| <b>RADIONUCLIDE: <span style="color: red;">Cl-36</span></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |                                    | <b>FORMS: Soluble</b>                                                                                                                                                                                                                                                                                                     |         |   |
| <b>PHYSICAL CHARACTERISTICS</b><br>HALF-LIFE: 300,000 years<br><b>DECAY EMISSIONS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |                                    |                                                                                                                                                                                                                                                                                                                           |         |   |
| Gammas / X-rays                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   | Betas / Positrons (+) / Electrons* |                                                                                                                                                                                                                                                                                                                           | Alphas  |   |
| E (keV)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | % | E (keV, Ave)                       | %                                                                                                                                                                                                                                                                                                                         | E (keV) | % |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   | 278                                | 98                                                                                                                                                                                                                                                                                                                        |         |   |
| - Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |                                    |                                                                                                                                                                                                                                                                                                                           |         |   |
| <b>STANFORD HAZARD CATEGORY</b><br>C – level (low hazard): $\leq 2$ mCi<br>B – level (moderate hazard): $> 2$ mCi, $\leq 100$ mCi<br>A – level (high hazard): $> 100$ mCi                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |                                    |                                                                                                                                                                                                                                                                                                                           |         |   |
| <b>EXTERNAL RADIATION HAZARDS</b><br>Gamma dose rate, point source at 1 ft, 1 mCi:<br><b>~0 mrem/h</b><br>Beta dose rate to skin, point source at 1 ft, 1 mCi:<br><b>400 mrem/h</b><br>Contamination skin dose, uniform deposit of 1 $\mu$ Ci per cm <sup>2</sup> :<br><b>6600 mrem/h</b>                                                                                                                                                                                                                                                                                                                       |   |                                    | <b>INTERNAL RADIATION HAZARDS</b><br>Annual Limit on Intake: <b>2000 <math>\mu</math>Ci</b> (Ingestion)<br><b>200 <math>\mu</math>Ci</b> (Inhalation)<br><br>The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE). |         |   |
| <b>SHIELDING</b><br><b>Gammas/X-rays:</b><br>N/A<br><br><b>Betas/electrons:</b><br><b>2 mm</b> of plastic will absorb all emissions.<br>Bremsstrahlung may be created and require additional shielding.                                                                                                                                                                                                                                                                                                                                                                                                         |   |                                    | <b>DOSIMETRY AND BIOASSAY REQS</b><br>Whole-body and finger-ring dosimeters are required for handling <b>5 mCi</b> or more, or <b>1 mCi amounts weekly</b> . Urine assays may be required after large spills or contaminations.                                                                                           |         |   |
| <b>SPECIAL PROBLEMS AND PRECAUTIONS:</b> <ol style="list-style-type: none"> <li>1. Recommended survey probe: <b>PGM</b></li> <li>2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often.</li> <li>3. Survey work areas before, during, and after work. Instrument and smear surveys are required.</li> <li>4. Segregate waste to those with half-lives of greater than <b>120 days (excluding H3 and/or C14)</b>.</li> <li>5. Limit soluble waste to the sewer to less than <b>10 <math>\mu</math>Ci/day</b> per lab.</li> </ol> |   |                                    |                                                                                                                                                                                                                                                                                                                           |         |   |

**References:**

- Delacroix, D., Guerre, J.P., Leblanc, P., Hickman, C. (2002). Radionuclide and Radiation Protection Data Handbook (2<sup>nd</sup> ed.). Ashford, Kent: Nuclear Technology Publishing.
- Johnson, T.E., Birk, B.K. (2012). Health Physics and Radiological Health (4<sup>th</sup> ed.). Baltimore, MD: Lippincott Williams & Wilkins.
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